



Metabolic alteration in obese diabetes rats upon treatment with *Centella asiatica* extract



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ARTICLE INFO

Article history:

Received 26 May 2015

Received in revised form

8 December 2015

Accepted 2 January 2016

Available online 13 January 2016

Keywords:

Centella asiatica

NMR-based metabolomics

Obese-diabetic rats

Urine metabolites

Serum metabolites

ABSTRACT

Ethnopharmacological relevance: ‘Pegaga’ is a traditional Malay remedy for a wide range of complaints. Among the ‘pegaga’, *Centella asiatica* has been used as a remedy for diabetes mellitus. Thus, we decided to validate this claim by evaluating the in vivo antidiabetic property of *C. asiatica* (CA) on T2DM rat model using the holistic ¹H NMR-based metabolomics approach.

Method: In this study, an obese diabetic (mimic of T2DM condition) animal model was developed using Sprague–Dawley rats fed with a high-fat diet and induced into diabetic condition by the treatment of a low dose of streptozotocin (STZ). The effect of *C. asiatica* extract on the experimental animals was followed based on the changes observed in the urinary and serum metabolites, measured by ¹H NMR of urine and blood samples collected over the test period.

Results: A long-term treatment of obese diabetic rats with CA extract could reverse the glucose and lipid levels, as well as the tricarboxylic acid cycle and amino acid metabolic disorders, back towards normal states. Biochemical analysis also showed an increase of insulin production in diabetic rats upon treatment of CA extract.

Conclusion: This study has provided evidence that clearly supported the traditional use of CA as a remedy for diabetes. NMR-based metabolomics was successfully applied to show that CA produced both anti-hyperglycemic and anti-hyperlipidemic effects on a rat model. In addition to increasing the insulin secretion, the CA extract also ameliorates the metabolic pathways affected in the induced diabetic rats. This study further revealed the potential usage of CA extract in managing diabetes mellitus and the results of this work may contribute towards the further understanding of the underlying molecular mechanism of this herbal remedy.

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1. Introduction

‘Pegaga’ is the name used for Pennywort among the Malaysian population, which refers to a group of ground creeping plants including those belonging to the genus *Centella* and *Hydrocotyle*.

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The plant is often consumed traditionally as fresh salad, with the belief that it would remedy sicknesses, such as dysentery, hyster-oepilepsy, leprosy, liver complaints, rheumatism, hemorrhoids, and dizziness (Burkill, 1966). Previously, we have reported that different ‘pegaga’ plants showed different metabolite profiles (Maulidiani et al., 2012). In addition to this, the study also revealed that *Centella asiatica* (CA) extract possessed the most potent antioxidant activity among the ‘pegaga’ species (Maulidiani et al., 2013). This finding and the notion that there is a close association between oxidative stress, inflammation and diabetes (Montane et al., 2014), encouraged us to investigate the antidiabetic property of this plant using metabolomics approach. This intention was

further substantiated by the claim that CA has also been used for the treatment of diabetes (Gayathri et al., 2011; Semenya et al., 2012). CA has been the most widely studied among the 'pegaga' plants from the phytochemical perspectives as well as on its biological activity. The terpenoid components of CA, including asiaticoside, asiatic acid, madecassoside and madecassic acid are commonly associated with the bioactivity characteristics of the plant (Gunther and Wagner, 1996).

Animal model has been generally used in the studies of diabetes. Non-genetic out-bred Sprague–Dawley rats are frequently used as the animal model by feeding them with a high-fat diet followed by diabetic induction by the treatment of a low dose (e.g., 30 mg/kg) of streptozotocin (STZ) (Srinivasan et al., 2005). This animal model mimics the typical symptoms of diabetic patients, such as hyperglycemia and dyslipidemia, characterized by the low level of high-density lipoprotein (HDL), and the high levels of low-density lipoprotein (LDL) and triglycerides (TG) (Srinivasan et al., 2005; Ramachandran et al., 2013).

Some considerable attempts have been conducted to evaluate the anti-diabetic property of CA extract and its bioactive compounds, e.g., asiatic acid. The treatment of the CA extract on STZ-induced diabetic rats revealed the anti-hyperglycemic and anti-hyperlipidemic activities of CA (Gayathri et al., 2011; Kabir et al., 2014). Concurrently, oral treatment of STZ-induced diabetic rats with asiatic acid was also found to lower their blood glucose levels, enhance insulin secretion via regeneration of β -cells, and modulation hepatic enzymes responsible for glucose metabolism (Liu et al., 2010; Ramachandran and Saravanan, 2013b). Moreover, asiatic acid has the ability to ameliorate oxidative stress in tissues of STZ-induced diabetic rats, as evidenced by the improved glycemic and antioxidant status along with decreased lipid peroxidation (Ramachandran and Saravanan, 2013a). Subsequently, Ramachandran et al. (2013) also showed that, in addition to its anti-diabetic effects, asiatic acid exhibited anti-hyperlipidemic effect in the experimental diabetic model. Despite all these reports, the molecular mechanism of the antidiabetic action of *C. asiatica* extracts still remains unclear.

Metabolomics refers to a comprehensive analysis of metabolites (i.e., low molecular weight molecules) in biological samples using a combination of spectral measurements such as NMR and mass spectroscopy, and multivariate statistical analysis, typically the PCA and PLS-DA methods (Nicholson et al., 1999). Numerous studies have shown that NMR-based metabolomics can be effectively utilized for identification and quantification of metabolites in the urine, serum or kidney tissue samples of diabetic rat models (Salek et al., 2007; Zhang et al., 2008b; Jankevics et al., 2009; Zhao et al., 2010). Moreover, NMR-based metabolomics has been shown to be a very efficient and robust method to evaluate the effects of thiazolidinediones (Van Doorn et al., 2007), repaglinide, metformin or rosiglitazone (Bao et al., 2009), oral triethylenetetramine (TETA) (Ugarte et al., 2012), Zhibai Dihuang pill (Zhao et al., 2012), and chinese medicine Gegen Qinlian decoction (Tian et al., 2013), in the treatment of diabetic cases. These studies point to the potential of metabolomics as a tool for identification the overall alteration in metabolite levels resulting from the treatments of diabetic cases.

In this study, an obese-diabetic animal model was developed using non-genetic out-bred Sprague–Dawley rats fed with high fat diet and induced into diabetic condition with a low-dose of STZ. Subsequently, ^1H NMR-based metabolomics was used to identify and compare the metabolic changes in the biofluids of the obese-diabetic rats upon treatment with *C. asiatica* extract. Through this study, we sought to evaluate the antidiabetic properties and to further understand the metabolic alterations that occur in the experimental animals upon oral treatment with the plant extract.

2. Materials and methods

2.1. Chemicals

Ethanol 70% (v/v), purchased from ChemPur (Karlsruhe, Germany), was used as the solvent for extraction. For the animal study, normal rat chow was supplied by Gold Coin, Kuala Lumpur, Malaysia. Streptozotocin (STZ) and sodium azide (NaN_3) were acquired from Merck, Darmstadt, Germany and carboxymethylcellulose from Sigma-Aldrich, St. Louis, US. The following chemicals were used for the NMR analysis: imidazole (Merck, Darmstadt, Germany), deuterated water (D_2O) (Sigma-Aldrich, St. Louis, US), trimethylsilylpropionic acid sodium salt (TSP) (Merck, Darmstadt, Germany), KH_2PO_4 (Merck, Darmstadt, Germany) and NaOD (Merck, Darmstadt, Germany).

2.2. Preparation of plant extracts

The leaves of CA were collected from University Agriculture Park, UPM, in mid-January 2011. The voucher specimen (no. SK1501/2008) was deposited at the herbarium of the Institute of Bioscience, Universiti Putra Malaysia. The leaves were properly washed and drained (at room temperature of 24–25 °C), and kept in –80 °C refrigerator for overnight. They were then freeze dried for at least 8 h. About 5 kg of the freeze dried leaves were ground to powder and macerated with 70% (v/v) ethanol followed by ultrasonication for 60 min. The extraction process was repeated three times and the combined extracts was filtered. The filtered extract was then evaporated using rotary evaporator and lyophilized to remove the remaining water to yield 1.3 kg of extract (26% of dried weight). The extracts were stored at –80 °C prior to study.

2.3. Identification and quantitative analysis of chemical markers in CA extract

The quantitative analysis of four common chemical markers for CA extract (asiaticoside, madecassoside, asiatic acid and madecassic acid) was performed using High-Performance Liquid Chromatography (JASCO, Japan) with UV detection set at 206 nm wavelength (See Supplementary Data). Calibration curve of the four compounds was based on four different concentrations (1000, 500, 250, 100 ppm) giving R^2 values of 0.99. In this study, 300 mg/kg body weight of the extract was given orally to the obese-diabetic rats. Based on the results of the quantitative analysis, the 300 mg of CA extract would contain 1.51 mg madecassoside, 2.19 mg asiaticoside, 0.7 mg madecassic acid and 0.13 mg asiatic acid. The ^1H -NMR metabolite finger print of the CA and other pegaga extracts had also been recorded and metabolomics analysis on these data has been published (Maulidiani et al., 2012). In addition, the relative quantification of other compounds such as dicaffeoyl quinic acids and flavonoids was also performed using HPLC-DAD analysis (Maulidiani et al., 2014). The results revealed the higher content of phenolic compounds in the forms of caffeoylquinic acid derivatives (0.119 mg/100 g of extract) and flavonoids (1.088 mg/100 g of extract) in the CA extract.

2.4. Animal study

All experimental procedures were approved by the Animal Care and Use Committee (ACUC), Faculty of Medicine and Health Science, Universiti Putra Malaysia (ACUC No: UPM/FPSK/PADS/BRUHH/00407). Thirty nine male Sprague–Dawley rats (weight range of 100–150 g) were purchased from Sapphire Enterprise Sdn Bhd. During acclimatization period (one week; room temperature; 12/12 h light/dark cycle), the rats were given normal rat chow (Gold Coin, Malaysia) containing 14% fat, 61% carbohydrate and

25% protein and distilled water, *ad libitum*. After acclimatization, the rats were randomly divided into 5 groups comprising:

1. Normal diet (nd) group ($n=10$)
2. Obese (ob) group ($n=10$)
3. Obese-diabetic (obdb) group ($n=5$)
4. Obdb group treated with 300 mg/kg CA extract (CA; $n=6$)
5. Obdb group treated with 300 mg/kg metformin (MTF; $n=8$)

The animal model for obese condition was developed according to a previous report (Jalil et al., 2008). The high-fat diet (HFD) was formulated as 49% fat from ghee (milk fat) and corn oil, 32% carbohydrate, and 19% protein. After three months on HFD, the obese rats were injected intraperitoneally (i.p.) with a low-dose of STZ (25 mg STZ/kg body weight) to induce diabetes. The STZ solution was prepared in 0.05 M citrate buffer solution (pH 4.5). The metabolic characteristics of rat fed with a high fat diet (HFD) followed by injection with low dose STZ has been reported to closely mimic the disease events of human Type 2 diabetes (Srinivasan et al., 2005). After three days of STZ treatment the plasma glucose levels of the experimental rats were measured with a glucometer (Roche, Mannheim, Germany) and the measurements were repeated on day 7 to analyze the diabetic status of the rats. The rats were considered diabetic when their basal plasma glucose levels achieved the concentrations of higher than 13.9 mmol/L (Liu et al., 2010). The diabetic animals were treated with the CA extract to assess its anti-diabetic effect and with metformin (MTF) to be used as control.

2.5. Administration of plants extracts to experimental animals

The CA extract and MTF were suspended in 0.03% (w/v) of carboxymethylcellulose (CMC) and administered daily (at 300 mg/kg) to the experimental animals by gastric intubation using a force-feeding needle. The treatment was continued over a period of 4 weeks. During this interval, rats in the control groups (normal, obese, and obese diabetes) received 0.03% (w/v) CMC.

2.6. Preparation of biofluid samples and acquisition of ^1H NMR spectra

Urine samples were collected from the control and treated groups at the initial, baseline, middle and final time points (TP). The initial TP of urine collections was at week 12 (three months after high fat diet regiment). Urine samples were also collected after induction of diabetes (week 13; baseline TP), after two weeks of treatment (week 15; middle TP), and after four weeks of treatment (week 17; final TP). According to Bollard et al. (2001), more metabolites can be detected in the urine of experimental animals collected overnight. The difference can be attributed to the fact that rats are nocturnal animals, and are therefore metabolically more active at night. Based on this information, the urine sample collections in this study were carried out overnight. Urine samples were collected in tubes containing 0.1% sodium azide, as an anti-microbial agent, by placing each rat in a metabolic cage and left overnight, before and after treatment. The collected urine samples were stored at -80°C prior to analysis.

For the ^1H NMR analysis, phosphate buffer (KH_2PO_4) was prepared in D_2O containing 0.1% TSP and 0.1% of imidazole. The pH of the D_2O was adjusted to pH 7.4 using 1 M NaOD solution. Thawed urine samples were centrifuged at 5000 rpm for 10 min. After this, 400 μl of the centrifuged urine sample was mixed with 200 μl phosphate buffer in a 1.5 ml Eppendorf tube and sonicated for 5 min at room temperature to remove air bubbles. The aliquots were then transferred into 5 mm NMR tubes. The ^1H -NMR data of the biofluids were obtained from a 500 MHz Varian INOVA NMR

spectrometer (Varian Inc., California, USA), functioning at a frequency of 499.887 MHz at room temperature (25°C). The ^1H NMR spectrum of each urine sample was recorded at 26°C (acquisition parameter: pulse width (PW) 21.0 μs (90°) and relaxation delay (RD) 2.0 s). For each sample, data of 64 scans were recorded with an acquisition time of 206 s. The spectral width was set to a range between -1.00 and 14.00 ppm. A PRESAT sequence was used to suppress residual water signal with low power selective irradiation. Deuterium oxide was used as the internal lock. A 0.1% of trimethylsilyl propionate (TSP) was used as a calibration standard.

The serum samples collected before and after treatment were also subjected to ^1H -NMR and biochemical analyses. Blood withdrawal of the rat was carried out by Retro-orbital sampling method. The procedure was conducted under general anesthesia, which involved penetrating the orbital sinus of rat's eye with a capillary tube (microhematocrit tube). The blood sample was collected in a vacutainer tube for serum collection (red topped tubes; without the addition of EDTA and anticoagulant). After collection, the whole blood was allowed to clot by leaving it undisturbed at room temperature for 15–30 min. The clotted blood was then centrifuged at 1000 g for 10 min, after which, the resulting supernatant (serum) was immediately transferred into a clean polypropylene tube (plain tube) using a sterile pipette. The serum samples were stored at -80°C prior to analysis.

The samples for ^1H NMR analysis of serum were prepared by mixing 200 μl of thawed serum sample with 400 μl of 0.01 M phosphate buffer (pH 7.4) containing 0.2% of TSP. Water signals and broad protein resonances were suppressed by the combination of PRESAT and the Carr–Purcell–Meiboom–Gill (CPMG) pulse sequence. The ^1H NMR spectra were measured at frequency 499.91 Hz with 128 scans to obtained 32,768 data points over the set spectral width (ranged from -1.00 to 14.00 ppm) and total acquisition time of 480 s.

2.7. Biochemical parameters

Biochemical analysis of serum samples was determined spectrometrically using a Chemistry Analyzer (Hitachi 902 Automatic Analyzer, Hitachi, Tokyo, Japan), and appropriate kits for the measurement of the typical serum parameters including glucose, cholesterol, LDL, and HDL. Plasma insulin level was determined using a radioimmunoassay (RIA) kit (Mercodia, Uppsala, Sweden).

2.8. Data processing and statistical analysis

The acquired NMR spectra were manually phased, baseline-corrected and referenced to TSP as an internal standard. The spectral region from 0.50 to 10.00 ppm was binned (0.04 ppm) using Chenomx NMR software (Chenomx NMR Suite 5.1 Professional, Edmonton, Canada). The signals at δH 4.68–5.00 (water), δH 5.55–6.00 (urea), and δH 7.23–7.40 and 8.43–8.46 ppm (imidazole) were excluded from the analysis. Data were Pareto-scaled and Principal Component Analysis (PCA) was performed using SIMCA-P software (version 12.0, Umetrics, Umea, Sweden). Statistical differences of metabolites from quantitative NMR spectra between groups were calculated using Tukey's honest significant difference (Tukey-HSD) multiple-comparison. The p -values of ≤ 0.05 were considered statistically significant.

3. Results

3.1. Biochemical analysis

Biochemical analyses of the serum from the obese diabetic rat group were conducted to confirm the success in establishing the

diabetic model (results included in [Supplementary data](#)). Before diabetic-induction with a low dose of STZ, the serum of obese group showed a significantly higher level of cholesterol than that of the normal group. After diabetic-induction, the glucose level of the obese diabetic group significantly increased ($p < 0.001$), and the insulin level decreased ($p < 0.05$). The results, as presented in [Supplementary data](#), showed that treatment of CA extract on diabetic rats lowered the glucose level significantly ($p < 0.001$). The serum insulin level for the CA-treated (46.49 pmol/L) increased and attained similar level as that of the normal group (44.91 pmol/L). In addition, the insulin level of the CA-treated group was significantly different ($p < 0.05$) from the level of the obese diabetic- (37.66 pmol/L) rats. As for the cholesterol, LDL and HDL, the levels are generally lower in the CA-treated group at 1.28 ± 0.33 , 0.39 ± 0.10 , and 0.94 ± 0.26 mmol/L, respectively,

which were comparable with that of the control normal group at 1.38 ± 0.45 , 0.36 ± 0.14 , and 0.90 ± 0.30 mmol/L, respectively ([Supplementary data](#)).

3.2. NMR spectral analysis

The representative ^1H NMR spectra of urine and serum samples from obese diabetic and normal rat, taken at the final time point, are shown in [Fig. 1A–D](#). The metabolites were identified and assigned based on comparison with the library data available from Chenomx NMR Suite 5.1 Professional, (Edmonton, Canada). The results were also tabulated in [Tables 1](#) and [2](#). Visually, the urine and serum spectra showed the elevation of glucose levels in the obese diabetic in comparison to the normal rats.

The overlay of the ^1H NMR spectra of serum samples before

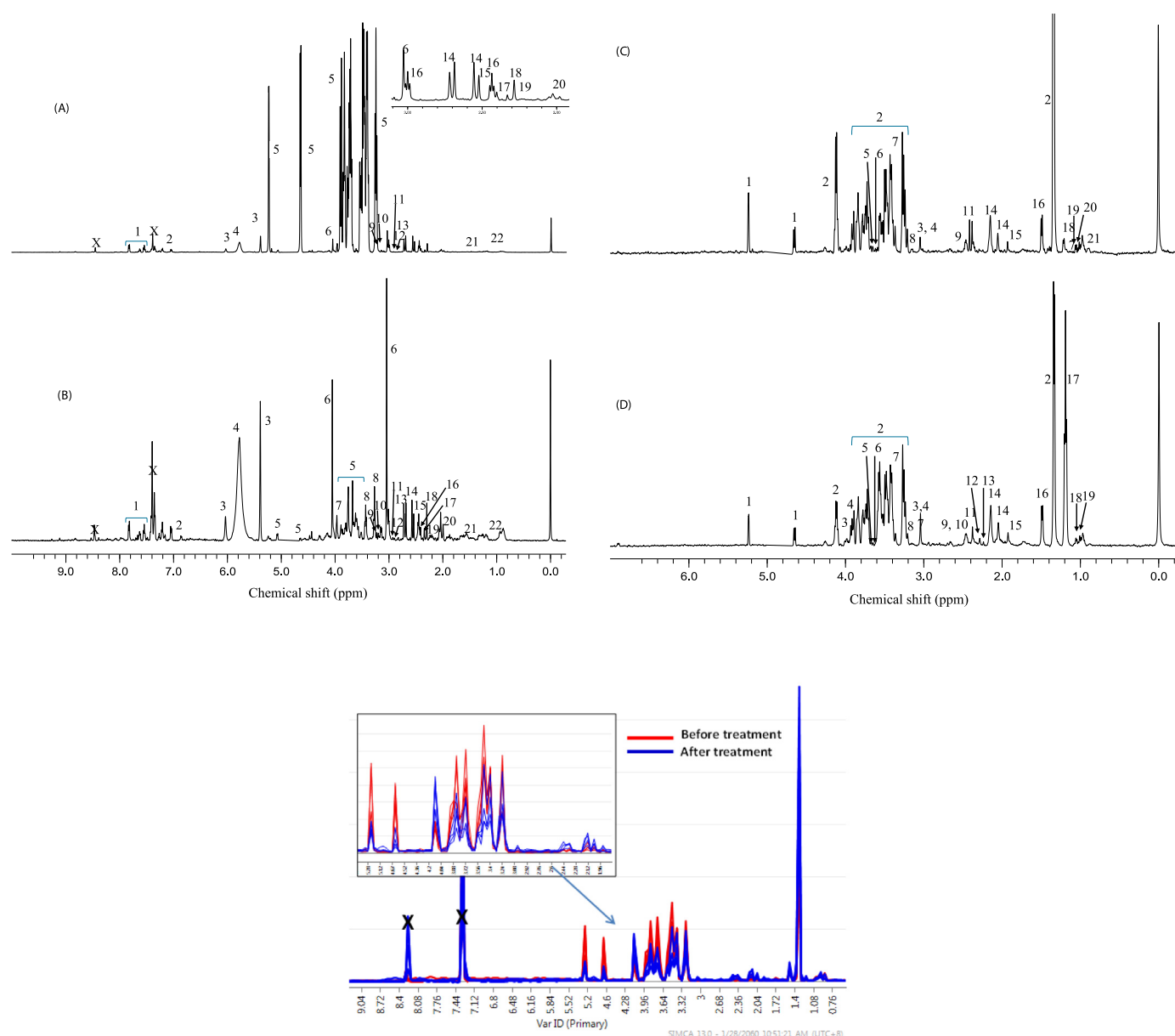


Fig. 1. ^1H NMR urine spectra for: (A) obese diabetic rat and (B) normal rat; and serum spectra for: (C) obese diabetic rat, and (D) normal rat groups; and (E) overlay of binned ^1H NMR data of obdb rats serum before and after treatment with CA extract. Identified urine metabolites: (1) hippurate; (2) aminohippurate; (3) allantoin; (4) urea; (5) glucose; (6) creatinine; (7) creatine; (8) taurine; (9) choline; (10) malonate; (11) dimethylglycine; (12) TMA; (13) DMA; (14) citrate; (15) 2-oxoglutarate; (16) succinate; (17) pyruvate; (18) acetoacetate; (19) acetone; (20) acetate; (21) alanine; and (22) 2-hydroxybutyrate. Identified serum metabolites: (1) glucose; (2) lactate; (3) creatinine; (4) creatine; (5) glycerol; (6) GPC; (7) TMAO; (8) choline; (9) glutamate; (10) succinate; (11) pyruvate; (12) acetoacetate; (13) acetone; (14) N-acetyl groups; (15) acetate; (16) alanine; (17) ethanol; (18) β -hydroxybutyrate; (19) isoleucine; (20) leucine; and (21) lipoprotein. (For interpretation of the references to color in this figure, the reader is referred to the web version of this article.)

Table 1
Metabolite changes in rat urine identified by ^1H NMR and their variations among groups and the respective p -values.

Metabolites	δH (ppm)	CA-treated					MTF-treated				
		Baseline	Middle	Final	Changes ^a	p^b	Baseline	Middle	Final	Changes ^a	p^b
Glucose	4.64	0.0613 $\pm$ 0.0057	0.0021 $\pm$ 0.0002	0.0022 $\pm$ 0.0012	–	< 0.001	0.0526 $\pm$ 0.0130	0.0029 $\pm$ 0.0013	0.0023 $\pm$ 0.0013	–	< 0.001
Glucose	5.24	0.0344 $\pm$ 0.0039	0.0033 $\pm$ 0.0021	0.0030 $\pm$ 0.0004	–	< 0.001	0.0311 $\pm$ 0.0068	0.0034 $\pm$ 0.0006	0.0022 $\pm$ 0.0005	–	< 0.001
Pyruvate	2.32	0.0004 $\pm$ 0.0003	0.0046 $\pm$ 0.0007	0.0052 $\pm$ 0.0016	+	< 0.001	0.0008 $\pm$ 0.0005	0.0047 $\pm$ 0.0016	0.0053 $\pm$ 0.0020	+	< 0.001
Lactate	1.32	0.0006 $\pm$ 0.0005	0.0033 $\pm$ 0.0005	0.0035 $\pm$ 0.0010	+	< 0.001	0.0010 $\pm$ 0.0006	0.0045 $\pm$ 0.0017	0.0036 $\pm$ 0.0007	+	< 0.001
Citrate	2.68	0.0021 $\pm$ 0.0014	0.0125 $\pm$ 0.0022	0.0197 $\pm$ 0.0104	+	< 0.001	0.0046 $\pm$ 0.0053	0.0050 $\pm$ 0.0046	0.0072 $\pm$ 0.0044	+	1
Fumarate	6.52	0.0002 $\pm$ 0.0001	0.0010 $\pm$ 0.0003	0.0013 $\pm$ 0.0003	+	< 0.001	0.0003 $\pm$ 0.0002	0.0007 $\pm$ 0.0002	0.0010 $\pm$ 0.0002	+	< 0.001
Succinate	2.40	0.0016 $\pm$ 0.0011	0.0069 $\pm$ 0.0010	0.0109 $\pm$ 0.0027	+	< 0.001	0.0035 $\pm$ 0.0034	0.0052 $\pm$ 0.0045	0.0042 $\pm$ 0.0016	+	1
2-oxoglutarate	2.44	0.0022 $\pm$ 0.0017	0.0135 $\pm$ 0.0027	0.0264 $\pm$ 0.0074	+	< 0.001	0.0061 $\pm$ 0.0067	0.0053 $\pm$ 0.0039	0.0060 $\pm$ 0.0028	–	1
β -Hydroxybutyrate	0.96	0.0005 $\pm$ 0.0004	0.0027 $\pm$ 0.0002	0.0030 $\pm$ 0.0007	+	< 0.001	0.0007 $\pm$ 0.0004	0.0023 $\pm$ 0.0006	0.0038 $\pm$ 0.0019	+	< 0.001
Acetoacetate	2.28	0.0004 $\pm$ 0.0003	0.0089 $\pm$ 0.0018	0.0102 $\pm$ 0.0014	+	< 0.001	0.0014 $\pm$ 0.0009	0.0056 $\pm$ 0.0021	0.0057 $\pm$ 0.0010	+	< 0.001
Acetone	2.24	0.0004 $\pm$ 0.0003	0.003 $\pm$ 0.0005	0.0050 $\pm$ 0.0014	+	< 0.001	0.0007 $\pm$ 0.0005	0.0026 $\pm$ 0.0007	0.0029 $\pm$ 0.0009	+	0.001
Acetate	1.96	0.0005 $\pm$ 0.0004	0.0036 $\pm$ 0.0004	0.0038 $\pm$ 0.0010	+	< 0.001	0.0011 $\pm$ 0.0008	0.0030 $\pm$ 0.0010	0.0031 $\pm$ 0.0012	+	0.004
Alanine	1.48	0.0004 $\pm$ 0.0003	0.0023 $\pm$ 0.0003	0.0020 $\pm$ 0.0006	+	< 0.001	0.0007 $\pm$ 0.0003	0.0014 $\pm$ 0.0005	0.0020 $\pm$ 0.0008	+	0.001
Hippurate	7.52	0.0008 $\pm$ 0.0006	0.0073 $\pm$ 0.0034	0.0147 $\pm$ 0.0066	+	< 0.001	0.0019 $\pm$ 0.0013	0.0056 $\pm$ 0.0020	0.0026 $\pm$ 0.0007	+	1
Dimethylamine	2.72	0.0015 $\pm$ 0.0008	0.0125 $\pm$ 0.0036	0.0152 $\pm$ 0.0069	+	< 0.001	0.0037 $\pm$ 0.0023	0.0077 $\pm$ 0.0017	0.0099 $\pm$ 0.0015	+	0.053
Creatinine	3.04	0.0070 $\pm$ 0.0063	0.0077 $\pm$ 0.0175	0.0552 $\pm$ 0.0183	+	< 0.001	0.0163 $\pm$ 0.0105	0.2363 $\pm$ 0.0580	0.1883 $\pm$ 0.0247	+	< 0.001
Creatinine	4.04	0.0045 $\pm$ 0.0035	0.0358 $\pm$ 0.0027	0.0330 $\pm$ 0.0030	+	< 0.001	0.0084 $\pm$ 0.0054	0.0240 $\pm$ 0.0069	0.0262 $\pm$ 0.0102	+	< 0.001
Trimethylamine	2.88	0.0002 $\pm$ 0.0001	0.0021 $\pm$ 0.0009	0.0014 $\pm$ 0.0004	+	0.001	0.0003 $\pm$ 0.0002	0.0019 $\pm$ 0.0010	0.0018 $\pm$ 0.0006	+	< 0.001
Allantoin	5.40	0.0040 $\pm$ 0.0026	0.0253 $\pm$ 0.0035	0.0206 $\pm$ 0.0056	+	1	0.0074 $\pm$ 0.0038	0.0232 $\pm$ 0.0038	0.0320 $\pm$ 0.0106	+	< 0.001

^a Changes is calculated from final urine samples subtracted by the baseline. Metabolites with “+/-” means increased/decreased.

^b Value of $p < 0.001$ is very significant, $p 0.05$ – 0.001 significant and $p > 0.05$ not significant.

Table 2Comparison of the metabolite levels in rat serum identified by ^1H NMR and their variations among groups and the respective p -values in the final time point.

Metabolites	δH (ppm)	Nd	Obdb	CA	obdb vs nd		CA vs obdb		CA vs nd	
					Changes ¹	p^2	Changes ¹	p^2	Changes ¹	p^2
Glucose	5.24	0.0162 $\pm$ 0.0022	0.0411 $\pm$ 0.0066	0.0170 $\pm$ 0.0027	+	< 0.001	–	< 0.001	+	0.939
Glucose	4.64	0.0101 $\pm$ 0.0035	0.0235 $\pm$ 0.0030	0.0063 $\pm$ 0.0023	+	< 0.001	–	< 0.001	–	0.123
Lactate	4.12	0.0174 $\pm$ 0.0025	0.0238 $\pm$ 0.0035	0.0402 $\pm$ 0.0037	+	0.007	+	< 0.001	+	< 0.001
Creatinine	4.04	0.0023 $\pm$ 0.0005	0.0018 $\pm$ 0.0008	0.0023 $\pm$ 0.0007	–	0.430	+	0.472	+	0.993
Creatine	3.92	0.0031 $\pm$ 0.0007	0.0025 $\pm$ 0.0010	0.0028 $\pm$ 0.0011	–	0.511	+	0.883	–	0.834
Glycerol	3.68	0.0069 $\pm$ 0.0011	0.0110 $\pm$ 0.0040	0.0072 $\pm$ 0.0026	+	0.034	–	0.099	+	0.977
GPC	3.64	0.0021 $\pm$ 0.0006	0.0032 $\pm$ 0.0013	0.0027 $\pm$ 0.0009	+	0.108	–	0.697	+	0.463
Ethanol	3.56	0.0397 $\pm$ 0.0087	0.0179 $\pm$ 0.0042	0.0118 $\pm$ 0.0015	–	< 0.001	–	0.392	–	< 0.001
Choline	3.20	0.0044 $\pm$ 0.0014	0.0042 $\pm$ 0.0018	0.0070 $\pm$ 0.0007	–	0.964	+	0.023	+	0.015
Creatine	3.04	0.0066 $\pm$ 0.0008	0.0019 $\pm$ 0.0012	0.0031 $\pm$ 0.0007	–	< 0.001	+	0.131	–	< 0.001
Glutamate	2.48	0.0030 $\pm$ 0.0009	0.0012 $\pm$ 0.0004	0.0018 $\pm$ 0.0007	–	0.003	+	0.518	–	0.040
Succinate	2.40	0.0034 $\pm$ 0.0023	0.0009 $\pm$ 0.0006	0.0055 $\pm$ 0.0029	–	0.169	+	0.020	+	0.281
Pyruvate	2.36	0.0046 $\pm$ 0.0013	0.0032 $\pm$ 0.0014	0.0050 $\pm$ 0.0010	–	0.17	+	0.133	+	0.891
Acetoacetate	2.28	0.0047 $\pm$ 0.0034	0.0010 $\pm$ 0.0007	0.0014 $\pm$ 0.0008	–	0.051	+	0.958	–	0.094
Acetone	2.20	0.0007 $\pm$ 0.0006	0.0004 $\pm$ 0.0002	0.0010 $\pm$ 0.0006	–	0.835	+	0.281	+	0.443
Glutamate	2.12	0.0119 $\pm$ 0.0015	0.0043 $\pm$ 0.0002	0.0076 $\pm$ 0.0029	–	< 0.001	+	0.075	–	0.007
Acetate	1.92	0.0047 $\pm$ 0.0010	0.0022 $\pm$ 0.0007	0.0029 $\pm$ 0.0009	–	0.001	+	0.554	–	0.009
Alanine	1.48	0.0138 $\pm$ 0.0033	0.0100 $\pm$ 0.0015	0.0133 $\pm$ 0.0029	–	0.088	+	0.220	–	0.963
Lactate	1.32	0.0893 $\pm$ 0.0220	0.1090 $\pm$ 0.0161	0.2232 $\pm$ 0.0451	+	0.494	+	< 0.001	+	< 0.001
Ethanol	1.20	0.0633 $\pm$ 0.0193	0.0021 $\pm$ 0.0007	0.0030 $\pm$ 0.0013	–	< 0.001	+	0.996	–	< 0.001
Leucin	1.00	0.0045 $\pm$ 0.0011	0.0041 $\pm$ 0.0007	0.0032 $\pm$ 0.0006	–	0.692	–	0.400	–	0.074
Isoleucin	0.92	0.0026 $\pm$ 0.0012	0.0021 $\pm$ 0.0009	0.0017 $\pm$ 0.0005	–	0.737	–	0.825	–	0.345
Lipoprotein	0.88	0.0045 $\pm$ 0.0020	0.0025 $\pm$ 0.0013	0.0034 $\pm$ 0.0017	+	0.153	+	0.754	–	0.513

¹ Metabolites with “+/-” means increased/decreased.² Value of $p < 0.001$ is very significant, $p 0.05$ – 0.001 significant and $p > 0.05$ not significant.

(red line) and after (blue line) the administration of CA extract obviously showed the decrease in glucose level (Fig. 1E). However, the changes in other metabolites that are present in lower concentrations were rather difficult to judge visually. Thus, further analysis based on the principal component analysis (PCA) was performed to identify and quantify the metabolite quantitative changes associated with CA extract and metformin treatments on the experimental obese diabetic animals.

3.3. Metabolites analysis of urine samples

A PCA-model was developed for the binned NMR spectra of urine samples collected from obese (HFD) rats before diabetic induction with STZ (initial TP) and after induction with STZ (baseline TP), as well as for the CA-, and MTF-treated obese-diabetic rats (at middle and final TPs). After auto-fitting, the PCA analysis resulted in nine components with good statistical performances ($R^2X=0.921$ and $Q^2=0.832$). The first principal component explains 70%, and the second 9.4% of the total variations. Therefore, the scores (t1/t2) and loadings (p1/p2) plots represent good summaries of the data set. For better visualization, the individual scores plots for the control i.e. normal (nd), obese (ob) and obese-diabetic (obdb) groups at their final TPs are presented in [Supplementary data](#), while the trajectory of metabolite changes from initial (red), baseline (green), middle (yellow), and final (blue) TPs, resulting from treatments with MTF and CA are shown in Fig. 2A and B, respectively.

The cluster migrations of the initial HFD-fed group (red dots) from +ve to -ve PC1 (baseline: green dots) after the STZ treatment, are clearly shown in Fig. 2A and B. The -ve PC1 is the region where the control obese-diabetic (obdb) rat group was also located ([Supplementary data](#)). This migration reflected the significant metabolic changes that occurred in the experimental obese animals upon receiving the low dose of STZ, confirming that the STZ treatment is successfully induced the groups into diabetic condition. Upon CA extract and MTF (Fig. 2A and B, respectively) treatments, the trajectories of cluster migrations were reverted,

back into the positive region of the PC1 (middle: yellow dots and final: blue dots). These clear migratory characteristics suggested that the long term treatment with CA extract was able to partially restore the metabolic state of the animal model towards normal conditions.

The loadings plot (Fig. 2C) implied that all negative p1 loadings have increased while the positive p1 loadings have decreased upon STZ treatment. Conversely, all positive p1 loadings have increased and all negative p1 loadings have decreased in the case of the obese-diabetic groups after the treatment with either CA extract or MTF (Fig. 2C). The plot suggested that: (1) only glucose was increased while the remaining metabolites were decreased in the obese-diabetic (obdb) group; and (2) the treatment of either CA extract or metformin decreased the glucose levels significantly ($p < 0.001$) and increased almost all the other metabolites (including pyruvate, lactate, dimethylamine, trimethylamine, acetoacetate, β -hydroxybutyrate, and creatinine), which were reduced in the obese-diabetic (obdb) condition. The TCA cycle metabolites, such as succinate, citrate, and 2-oxoglutarate, were increased significantly ($p < 0.001$) only by the treatment of CA extract, while they were only insignificantly increased ($p < 0.001$) by the treatment of MTF. These changes are summarized in [Table 1](#).

3.4. Metabolite analysis of serum samples

The PCA-model fitted to the binned NMR spectra of the serum from the control normal (nd: red dots), obese-diabetic (obdb: orange dots) and the CA-treated groups at baseline and final TPs (CABase: green dots, CAFin: blue dots) is shown in Fig. 3A. Four components were obtained using Auto-fit with $R^2X=0.792$ and $Q^2=0.596$, which indicated that the model was good. The first two PCs accounted for 64.4% of the variations and therefore, the t1/t2 and p1/p2 plots represented good summaries of the data set.

The control normal (nd) and obese-diabetic (obdb) groups were distinctly separated in the first principal component, with the normal group being located in negative t1, and obese-diabetic group in positive t1. Comparison of the metabolites between the

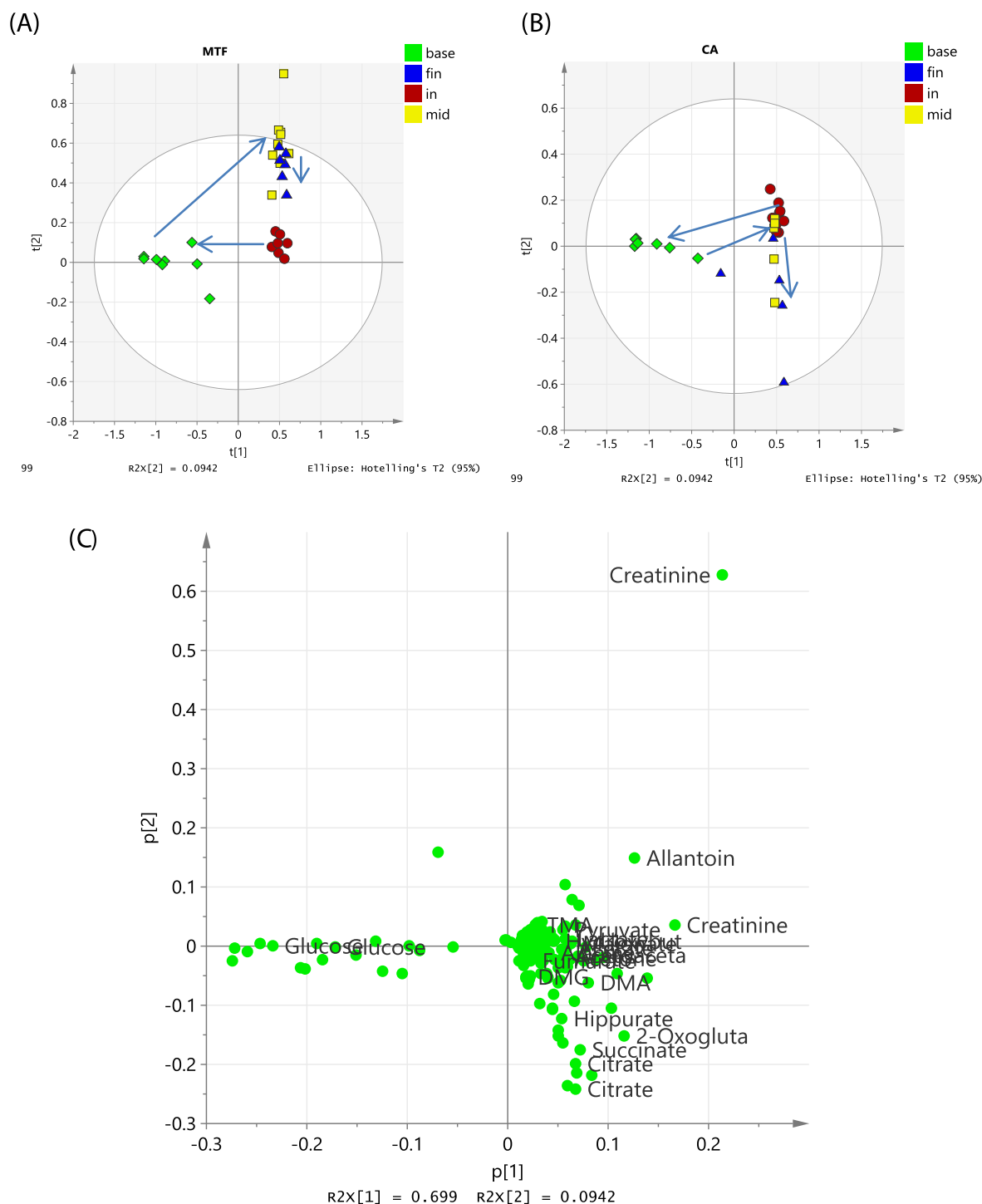


Fig. 2. (A) PCA scores plot for metformin treated, with trajectory; (B) PCA scores plot for *C. asiatica*, (CA)-treated, with trajectory; and (C) PCA loadings plot of the ^1H NMR data. (For interpretation of the references to color in this figure, the reader is referred to the web version of this article.)

obese-diabetic and normal groups indicated that the glucose, lactate, glycerol, glycerophosphorylcholine (GPC), and lactate levels increased in the obese-diabetic group while other metabolites, such as glutamate, ethanol, and acetate were significantly lower ($p < 0.05$). Other metabolites including creatinine, choline, succinate, pyruvate, acetoacetate, acetone, glutamate, alanine, ethanol, leucine, isoleucine, and lipoprotein were insignificantly lower, as observed from the loadings plot (Fig. 3B). After the treatment of CA, the glucose level was significantly decreased, whereas choline, succinate, and lactate were significantly elevated. Some of the metabolites, such as pyruvate, glutamate, alanine

were found to be increased but not significantly ($p < 0.05$). Glycerol, GPC, leucine and isoleucine were insignificantly decreased (Table 2). Comparison of the metabolite profiles of the normal, obese-diabetic and CA-treated groups, and the p -values of the detected metabolites in the serum are summarized in Table 2. Our findings, which showed the lowering of lipoprotein levels in obese-diabetic rats in comparison to the normal rats, however, is inconsistent with the current knowledge on diabetic conditions. Similar results have also been reported (Tian et al., 2013), which was rationalized by the possibility of precipitation (of lipoprotein) during plasma sample treatment. The migration of the CA-treated

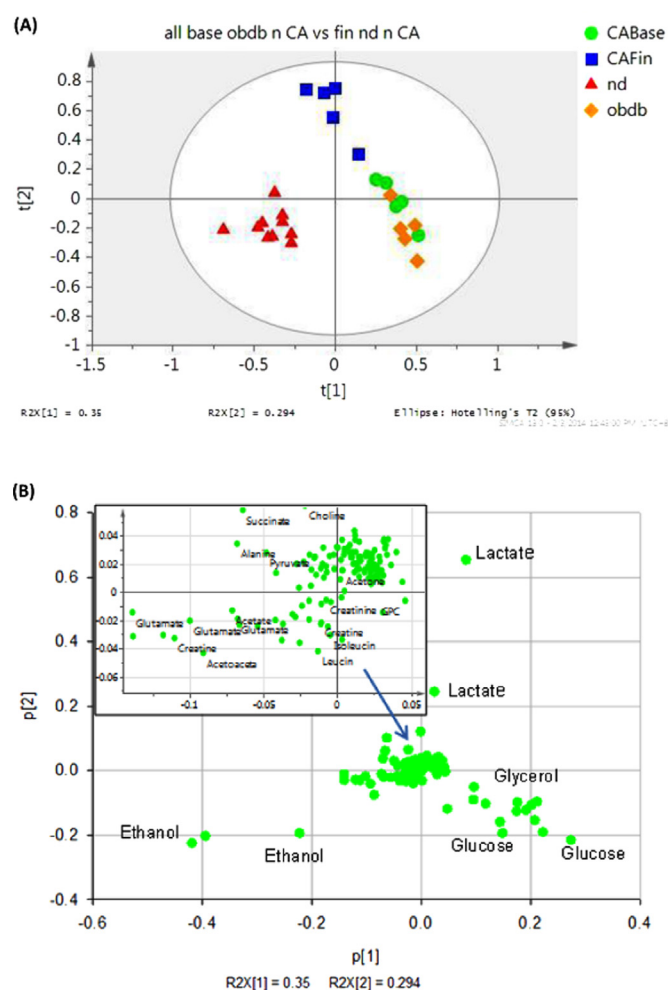


Fig. 3. (A) PCA scores data collected baseline and final TPs of the CA-treated diabetic, the normal diet and the obese diabetic rats, and (B) loadings plot of the changes in serum metabolites of the same groups. (For interpretation of the references to color in this figure, the reader is referred to the web version of this article.)

group scores, which showed a gradual reversal of the obese-diabetic towards normal state, again indicated that long term treatment with CA extract partially ameliorated the metabolic disturbances in diabetic condition, especially those related to TCA, glucose and lactate metabolisms.

4. Discussion

Our study showed that treatment of CA extract on obese diabetic rats can restore the disturbed metabolism back towards the normal condition. After 4 weeks of treatment, the glucose level was decreased, which agreed well with the previous findings (Gayathri et al., 2011; Kabir et al., 2014).

The basic mechanisms of anti-diabetic drugs include the improvement of insulin secretion, reduction of insulin resistance, as well as inhibition of carbohydrate digestion and glucose absorption (Krentz and Bailey, 2005). Previously, Kabir et al. (2014) concluded that CA extract manifested its anti-hyperglycemic activity by carbohydrate inhibition and glucose-fiber binding. The marker triterpenes in CA extract could be responsible for enhancing insulin secretion in this study, acting like glibenclamide (Ramachandran and Saravanan, 2013b, 2013a; Ramachandran et al., 2013). Jang et al. (2010) reported that triterpenoids have potential to inhibit ATP sensitive K^+ channels which leads to high insulin

secretion and consequently decreasing the blood glucose level.

Energy metabolism (i.e., glucose uptake and utilization, and TCA cycle) is also disturbed in diabetic condition. In Type 2 diabetes, the high gluconeogenesis [e.g. enhanced endogenous glucose production (EGP)] and low glycogenolysis increase the glucose level. Increasing glucose level inhibits three regulated enzymes in glycolysis, namely the hexokinase, phosphofructokinase, and pyruvate kinase (Belfiore et al., 1989; Vats et al., 2003), which can lead to the lowering of pyruvate and alanine levels observed in diabetic rats. Moreover, the TCA intermediates, such as citrate, fumarate, succinate, and oxoglutarate, are shown to be decreased in the diabetic rats, which could be associated to mitochondrial dysfunction. Our results showed that treatment of CA extract increased the pyruvate, alanine, lactate and the intermediates of TCA levels, thus indicating the increase in glycolysis, decrease in gluconeogenesis, and enhancement of the TCA cycle. In a normal condition, TCA cycle yields gluconeogenesis precursor and facilitate glucose oxidation to reinstate glucose homeostasis. Ramachandran and Saravanan (2013b) found that administration of asiatic acid to STZ-induced diabetic rats significantly reversed the activities of key enzymes, including hexokinase, glucose-6-phosphate dehydrogenase (G6PDH), and pyruvate kinase (PK), which elevate glycolysis.

Dyslipidemia is a condition characterized by the low level of high-density lipoprotein (HDL), and the high levels of low-density lipoprotein (LDL) and triglycerides (TG) in the serum of diabetic patients, which could eventually increase the risk of cardiovascular diseases (Ansarullah et al., 2011). In this study, the increased glycerol level in diabetic rats suggested an increase in lipid metabolism and energy metabolic shift from glucose to lipids. This finding agreed well with the previous results (Zhang et al., 2008a; Mooradian, 2009; Tian et al., 2013). Our results showed that treatment of CA extract decreased the levels of glycerol in serum, which indicates a decreased lipid metabolism. More recently, Ramachandran et al. (2013) showed that administration of asiatic acid in diabetic rats significantly increased the activities of lecithin cholesterol acyltransferase (LCAT) and plasma lipoprotein lipase (LPL), while decreasing the activity of HMG-CoA reductase. In relevance to this, the decrease in lipid metabolism exhibited by CA extract observed in our study could also be affected by the same mechanism.

Creatinine, a degradation product of creatine, is biosynthesized from arginine and glycine in the muscles (Walker, 1961; Wyss and Kaddurah-Daouk, 2000; Brosnan and Brosnan, 2010). In this study, the urine and serum creatinine levels in diabetic rats were lower in comparison to normal rats. This is consistent with the low urine creatinine level in Type 2 diabetic patients (Salek et al., 2007; Kumar et al., 2013). The levels of amino acids (leucine, isoleucine, alanine and glutamate) in serum of diabetic rats were also decreased, indicating an increase in protein metabolism through gluconeogenesis, which is a common phenomenon in diabetic condition.

Choline can be synthesized by the body as well as acquired through food consumption. Dumas et al. (2006) showed that choline is degraded into either methylamines [dimethylamine (DMA), trimethylamine N-oxide (TMAO), and trimethylamine (TMA)] by gut microbiota, or to creatinine through dimethylglycine in mammalia. They also showed that choline was used for the synthesis of phosphatidylcholine and GPC in both the bacteria and mammalia. Therefore, the disturbed gut bacterial activity, as well as the changed choline degradation, might lead to the decreased level of methylamines in diabetic rats. In addition, our study also revealed the increase of hippurate level in the CA treated rats. It has been suggested that the concentrations of aromatic metabolites e.g. hippurate in the urine, are closely related to the diet (Bell et al., 1991). Thus, the consumption of polyphenolic compounds

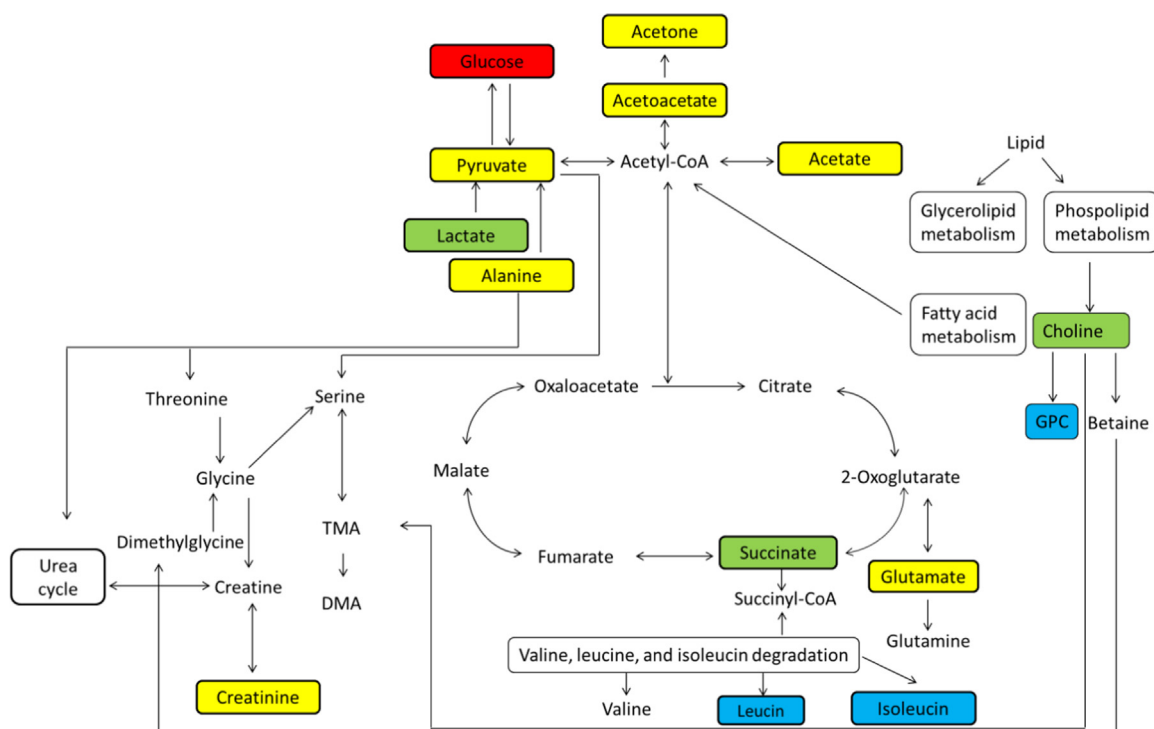


Fig. 4. Metabolic pathway alteration of the serum from CA-treated obese diabetic rats. The changes of metabolites after the treatment are shown in red (significantly decreased), blue (not significantly decreased), green (significantly increased), and yellow (not significantly increased), as compared to control obese diabetics. ($p < 0.05$ is considered to be significant while $p > 0.05$ is not significant). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

(e.g. flavonoids, phenolic acids) may result in high hippurate content (Lees et al., 2013). This situation may apply for the high hippurate content occurring in the CA treated rats group in our study, due to the presence of flavonoids and phenolic acids components in the extract. The metabolic alteration taking place in CA treated obese diabetic rat is represented in Fig. 4.

5. Conclusions

The NMR-based metabolomics analysis on the urine and serum samples has been successfully used in identifying and quantifying metabolite changes in diabetic pathogenesis and the effects of CA extract in an obese-diabetic animal model. The current study showed that long term treatment of CA extract on obese diabetic condition could reverse glucose, lipid, tricarboxylic acid cycle, and amino acid metabolic disorders back towards the normal state. The results from biochemical analysis also suggested that CA extract ameliorates insulin secretion. On account of these results, the antidiabetic property exhibited by CA extract could involve enhancement of insulin secretion resulting from β -cell inducement by asiatic acid, and amelioration of specific metabolic pathways. The results also confirmed the traditional claim that CA extract has anti-hyperglycemic and anti-hyperlipidemic effects, and therefore, could be useful for the treatment of diabetes mellitus.

^1H NMR-based metabolomics analysis of both the urine and serum is an important source of information on metabolic fingerprint data. The peaks of several key metabolites are well resolved in the urine NMR spectra (e. g. TCA compounds, hippurate, creatinine and allantoin), while others, such as lactate, as well as trace biomarkers of lipid and amino acid metabolisms (eg. glycerol, GCP, alanine, leucine, isoleucine), can be identified easily in the serum NMR spectra. For experiments requiring multiple collections over time space, the application of urine NMR-based metabolomics is the most practical method due to its relative ease of

collection, handling of the sample, and shorter experiment runtime.

Acknowledgments

We would like to thank the Ministry of Higher Education Malaysia for supporting this study through Fundamental Research Grant Scheme (vote no.: 5524044).

Appendix A. Supplementary material

Supplementary data associated with this article can be found in the online version at <http://dx.doi.org/10.1016/j.jep.2016.01.001>.

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